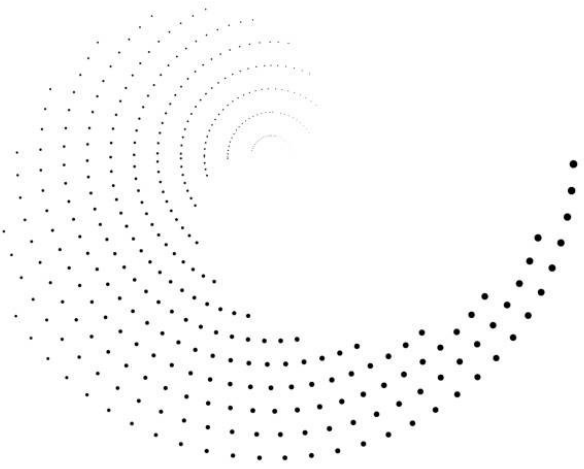




Operating system and system programming (25CS2104E)

LINUX PROCESS MONITORING AND CONTROL SYSTEM

Review-I

Under the Esteemed Guidance of

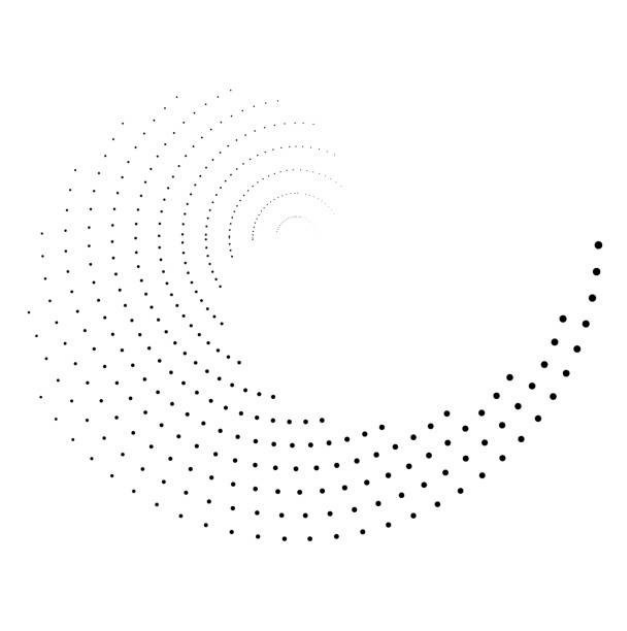
Raghupathi M
CSE Department

Team Members:

P.Ruhitha 2520080053

M.Ananya 2520080069

Ch.Madhulika 2520090056



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Introduction

A Linux operating system may have many processes running simultaneously, including user applications and background system tasks. The operating system is responsible for allocating resources to these processes and managing their execution efficiently.

Each process has important attributes such as a Process ID (PID), process state, parent process, and memory usage. Linux provides mechanisms that allow this information to be accessed and processes to be managed when necessary.

Understanding these mechanisms gives us a practical view of how the operating system manages processes and how user programs interact with the Linux kernel. Our project focuses on exploring these concepts through a simple Linux-based process monitoring and control system.



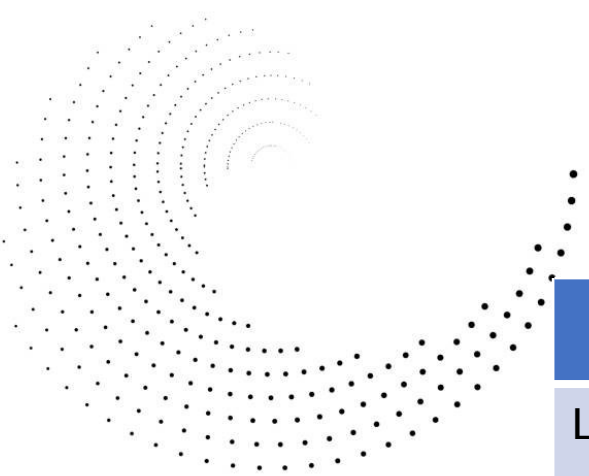
Abstract

Modern operating systems run multiple processes simultaneously, making it important to monitor and manage these processes efficiently. This project proposes a Linux-based Process Monitoring and Control System that allows users to view information about running processes and perform basic process management operations through a command-line interface.

The system will be developed using the C programming language and Linux/POSIX mechanisms. It will obtain process information such as Process ID (PID), process name, process state, parent process, and memory usage using the Linux /proc filesystem. The system will also provide basic process control operations such as pausing, resuming, and terminating a selected process using Linux signals.

The project focuses on understanding fundamental Operating Systems concepts including process management, process states, process identification, signals, system calls, and interaction with the Linux kernel. The proposed solution will provide a simple and practical way to observe how processes are created, monitored, and controlled in a Linux environment.

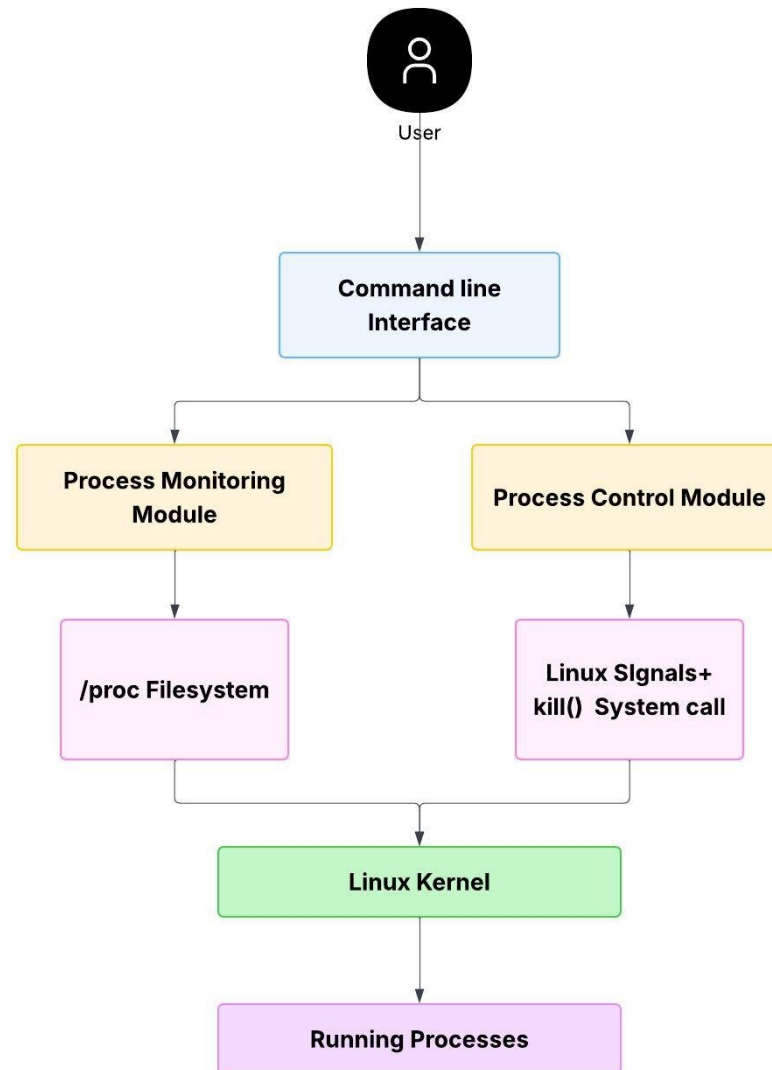
The expected outcome is a functional command-line application that displays process information and allows authorized users to perform basic process control operations. The project will help demonstrate Operating Systems concepts through practical Linux system programming rather than through a conventional graphical or web application.

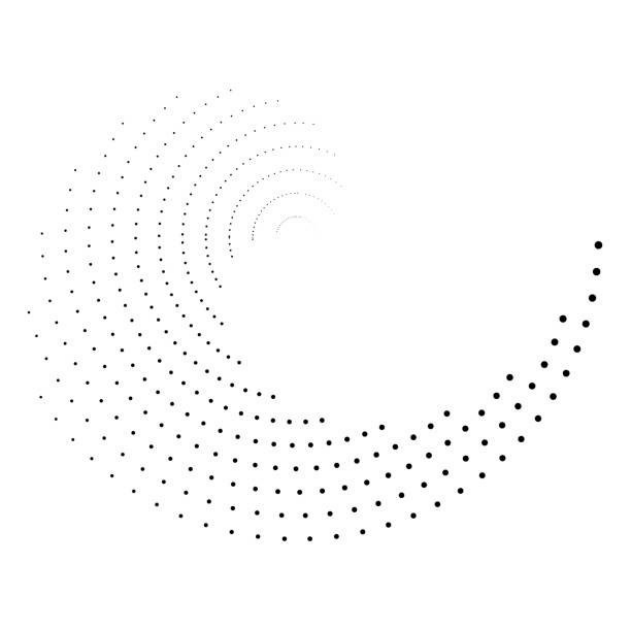


Hardware and Software requirements

SOFTWARE REQUIRED	HARDWARE REQUIRED
Linux/Ubuntu	Processor
C Programming language	RAM
GCC	Storage
POSIX APIs	Display
Linux /proc Filesystem	
Git & Github	

System Architecture





Thank you